

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 31-08-2026

1. TITLE

Design of a Basic Webpage using HTML Tags

2. AIM

To develop a webpage displaying college information utilizing fundamental HTML tags such as headings, paragraphs, links, lists, tables, images, forms, and multimedia elements.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML document.
- To implement text formatting, hyperlinks, and multimedia tags.
- To organize tabular data using HTML tables.
- To build a user input form for student registration.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language for creating web pages. It provides structural semantics for text and media. Elements like <header>, <nav>, <section>, <article>, and <footer> offer semantic meaning, while tags like <table>, , <audio>, and <video> enable embedding of data and media. Forms allow collection of user input via various input types.

5. ALGORITHM / PROCEDURE

1. Create index.html and define the basic document structure.
2. Add a header and navigation links to different sections.
3. Use text formatting tags (bold, italic, sub, sup) in the about section.
4. Implement ordered and unordered lists for courses and facilities.
5. Add internal and external hyperlinks using the anchor tag.
6. Design a student details table with borders and cell padding.
7. Embed images for the college logo and campus.
8. Create a registration form with various input types.
9. Add audio and video controls for multimedia playback.
10. Test and verify the page structure in a browser.

6. CODE EXPLANATION

The HTML document starts with the `<!DOCTYPE html>` declaration, specifying it as an HTML5 document. Semantic tags such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` are used to structure the page content logically. Paragraphs (`<p>`), text formatting tags (`<b>`, `<i>`), ordered (`<ol>`), and unordered (`<ul>`) lists are implemented to display structured text, course offerings, and college facilities. Hyperlinks (`<a>`) provide navigation to external resources. Tabular data is organized using the `<table>` element with custom borders and cell padding. User input is collected using interactive form elements (`<form>`, `<input>`, `<select>`, `<textarea>`) supporting various HTML5 input types. Lastly, multimedia elements like `<audio>` and `<video>` are embedded to provide audio-visual controls directly within the webpage.

7. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>
<!-- PART A : BASIC HTML -->
  <!-- Heading -->
  <h1>College Information Portal</h1>
  <h2>Anil Neerukonda Institute of Technology and Sciences (ANITS)</h2>
  <!-- Paragraph -->
  <p>
    <b>Anil Neerukonda Institute of Technology and Sciences (ANITS)</b> is one of the
    leading engineering colleges in <i>Visakhapatnam, Andhra Pradesh</i>.
    The college was established in 2001 and is affiliated with
    Andhra University. It offers various undergraduate and postgraduate
    programs in engineering and technology.
  </p>
  <hr>
  <!-- Ordered List -->
  <h3>Courses Offered</h3>
  <ol>
    <li>Computer Science and Engineering (CSE)</li>
    <li>Computer Science and Engineering (CSM)</li>
    <li>Electronics and Communication Engineering (ECE)</li>
    <li>Electrical and Electronics Engineering (EEE)</li>
    <li>Mechanical Engineering</li>
    <li>Civil Engineering</li>
  </ol>
  <!-- Unordered List -->
  <h3>Facilities Available</h3>
  <ul>
    <li>Central Library</li>
    <li>Computer Laboratories</li>
    <li>Sports Complex</li>
    <li>Hostel Facility</li>
    <li>Wi-Fi Campus</li>
    <li>Cafeteria</li>
  </ul>
  <hr>
  <!-- Hyperlinks -->
  <h3>Useful Links</h3>
  <p>
```

```

        Visit the
        <a href="https://anits.edu.in" target="_blank">
            ANITS Official Website
        </a>
    </p>
<!-- PART B : TABLES & IMAGES -->
<section>
<h2>Student Details</h2>
<table border="1" cellpadding="10" cellspacing="0">
    <tr>
        <th>Roll No</th>
        <th>Name</th>
        <th>Branch</th>
        <th>Email</th>
    </tr>
    <tr>
        <td>A24126552100</td>
        <td>Nimidithalli Nandu</td>
        <td>CSM</td>
        <td>nandu@email.com</td>
    </tr>
    <tr>
        <td>A24126552101</td>
        <td>dinesh</td>
        <td>mech</td>
        <td>dinesh@gmail.com</td>
    </tr>
    <tr>
        <td>A24126552119</td>
        <td>sujetha</td>
        <td>ECE</td>
        <td>suraj@gmail.com</td>
    </tr>
</table>
<h2>College Logo</h2>

</section>
<!-- PART C : STUDENT FORM -->
<section>
<h2>Student Registration Form</h2>
<form>
<label>Name:</label><br>
<input type="text" required><br><br>
<label>Email:</label><br>
<input type="email" required><br><br>
<label>Password:</label><br>
<input type="password" required><br><br>
<label>Gender:</label><br>
<input type="radio" name="gender"> Male
<input type="radio" name="gender"> Female
<input type="radio" name="gender"> Other
<br><br>
<label>Hobbies:</label><br>
<input type="checkbox"> Reading
<input type="checkbox"> Sports

```

```

<input type="checkbox"> Music
<input type="checkbox"> Coding
<br><br>
<label>Branch:</label><br>
<select>
<option>CSE</option>
<option>CSM</option>
<option>ECE</option>
<option>EEE</option>
<option>Mechanical</option>
<option>Civil</option>
</select>
<br><br>
<label>Comments:</label><br>
<textarea rows="5" cols="40"></textarea>
<br><br>
<input type="submit" value="Submit">
<input type="reset" value="Reset">
</form>
</section>
<!-- PART D HTML FEATURES -->
<header>
<h1>ANITS College Information Portal</h1>
</header>
<nav>
<a href="#">Home</a> |
<a href="#">Departments</a> |
<a href="#">Admissions</a> |
<a href="#">Contact</a>
</nav>
<section>
<article>
<h2>About ANITS</h2>
<p>
ANITS is one of the leading engineering colleges in
Visakhapatnam offering quality education in engineering,
technology and management.
</p>
</article>
</section>
<h2>Audio</h2>
<audio controls>
<source src="college_song.mp3" type="audio/mpeg">
Your browser does not support audio.
</audio>
<br><br>
<h2>Video</h2>
<video width="500" controls>
<source src="college_video.mp4" type="video/mp4">
Your browser does not support video.
</video>
<br><br>
<h2>HTML5 Input Types</h2>
<form>
<label>Email:</label>
<input type="email">

```

```
<br><br>
<label>Date of Birth:</label>
<input type="date">
<br><br>
<label>Age:</label>
<input type="number">
<br><br>
<label>Favorite Color:</label>
<input type="color">
<br><br>
<label>Skill Level:</label>
<input type="range" min="0" max="100">
</form>
<footer>
<hr>
<p> 2026 ANITS College Information Portal. All Rights Reserved.</p>
</footer>
</body>
</html>
```